IMAGE PROCESSING & COMPUTER VISION

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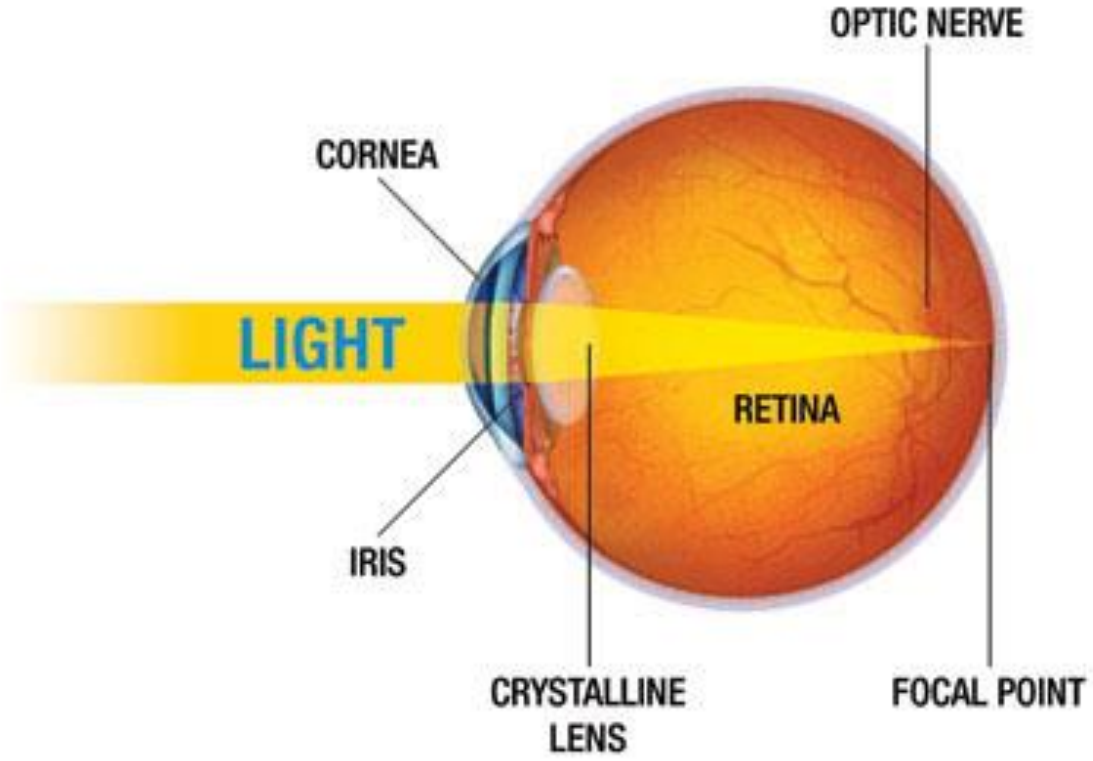
CEDRE 2026

FUNDAMENTALS

Chapter 1

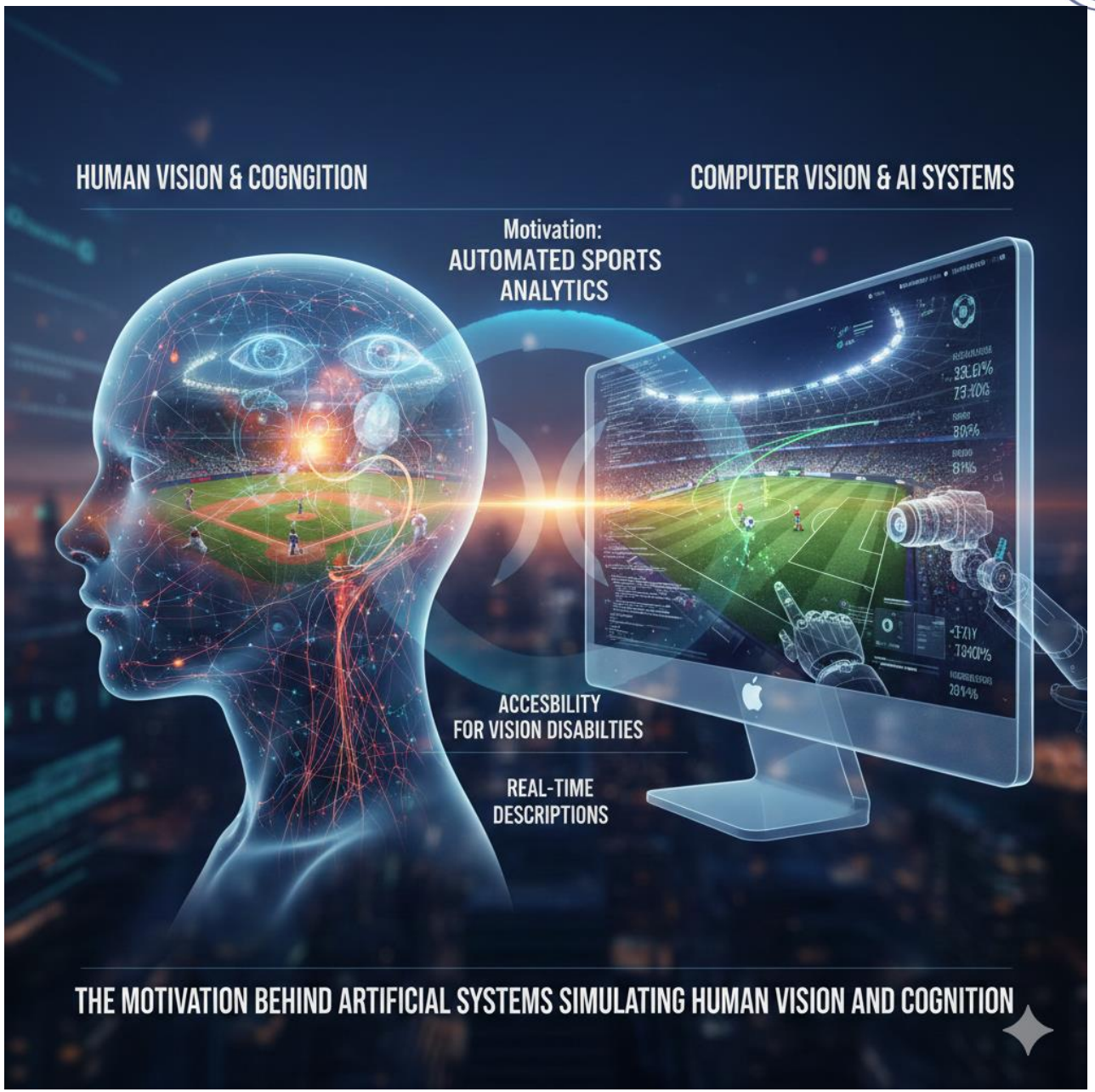
VISION


OPTIC NERVE
CORNEA
LIGHT
RETINA
IRIS
CRYSTALLINE LENS
FOCAL POINT

Most of us know that sunlight is responsible for sustaining life on our planet, but have you ever wondered how this shaped our lives? For starters, almost every being on the earth has some way to sense it (even some bacteria and single-celled organisms). Humans share this ability, but we have a much more complicated system for interacting with light. We capture light through a lens that then emits an electrical signal in our eyes that passes through a cable-like structure (our nervous system), and that signal gets reconstructed to tell us what our surroundings look like in our brains.

This process is what we call vision. It is a fundamental step in our evolution. It is so important that scientists have hypothesized that the development of centralized nervous systems (which ultimately lead us to our big brains) follows the advent of vision. It makes sense without sensors capturing such vast information, why waste resources on making the machinery required to develop it?

Mr.


HUMAN VISION & COGNGITION
COMPUTER VISION & AI SYSTEMS
Motivation:
AUTOMATED SPORTS
ANALYTICS
ACCESBILITY
FOR VISION DISABILTIES
REAL-TIME
DESCRIPTIONS
THE MOTIVATION BEHIND ARTIFICIAL SYSTEMS SIMULATING HUMAN VISION AND COGNITION

If you ever spontaneously kicked a ball, your brain performs a myriad of tasks unconsciously in a split second. It correctly identifies the ball, tracks its movement, predicts its trajectory, calculates the speed at which the ball would arrive at your location, predicts your foot trajectory, adjusts the strength and angle of impact, and sends the signal from your brain to your foot to change its position. To take an image as an input (in this case, the signal captured by our retina) and transform it into information (kicking the ball) is the core of computer vision. We will go into more detail about this in the next chapter.

Shockingly, we don't need any formal education for this. We don't attend classes for most of the decisions we make daily. No mental math 101 can estimate the foot strength required for kicking a ball. We learned that from trial and error growing up. And some of us might never have learned at all. This is a striking contrast to the way we built programs. Programs are mostly rule-based.

Let’s try to replicate just the first task that our brain did: detecting that there is a ball. One way to do it is to define what a ball is and then exhaustively search for one in the image. Defining what a ball is is actually difficult. Balls can be as small as tennis balls but as big as Zorb balls, so size won’t help us much. We could try to describe its shape, but some balls, like rugby, are not always perfectly spherical. Not everything spherical is a ball either, otherwise bubbles, candies, and even our planet would all be considered balls.

The Motivation Behind Creating Artificial Systems Capable of Simulating Human Vision and Cognition

Although they have similar input and output, human vision and computer vision are different processes. Sometimes they overlap. However, computer vision is primarily concerned with developing and understanding algorithms and models in vision systems and their decisions. It is not constrained to the creation of systems that replicate human vision. It can be used for problems that would be too tedious, time-consuming, expensive, or error-prone for humans to do. Our ball example is still a simple one, and you might not think that is super useful. However, a model capable of tracking a ball can be used in sports events to provide faster and more fair decisions during gameplay. With the popularization of image-to-text and text-to-speech models, we could also make live sports events more accessible for people who have vision disabilities by automatically tracking the ball and its players and describing it in real time. Thus, even simple use cases can have a positive impact on society.

IMAGE

- A visual representation of an object, a scene, a person, or even a concept.
- A function F(x,y)

The brightness at position (x, y)
Every point on the image has a position like : (x,y)

What is **F(x, y)**?
This is the **intensity (gray level)** at that position.

- F(10,20)=0 → black
- F(10,20)=255 → white
- F(10,20)=120 → gray

RGB image

$$f(x,y) = \begin{bmatrix} r(x,y) \\ g(x,y) \\ b(x,y) \end{bmatrix}$$

WHAT IS A **PIXEL**?

- One tiny square in the image at a specific (x, y) position
- Pixel = **position + intensity**
- Pixel at (3, 5) has intensity 200 ➔ F(3,5)=200

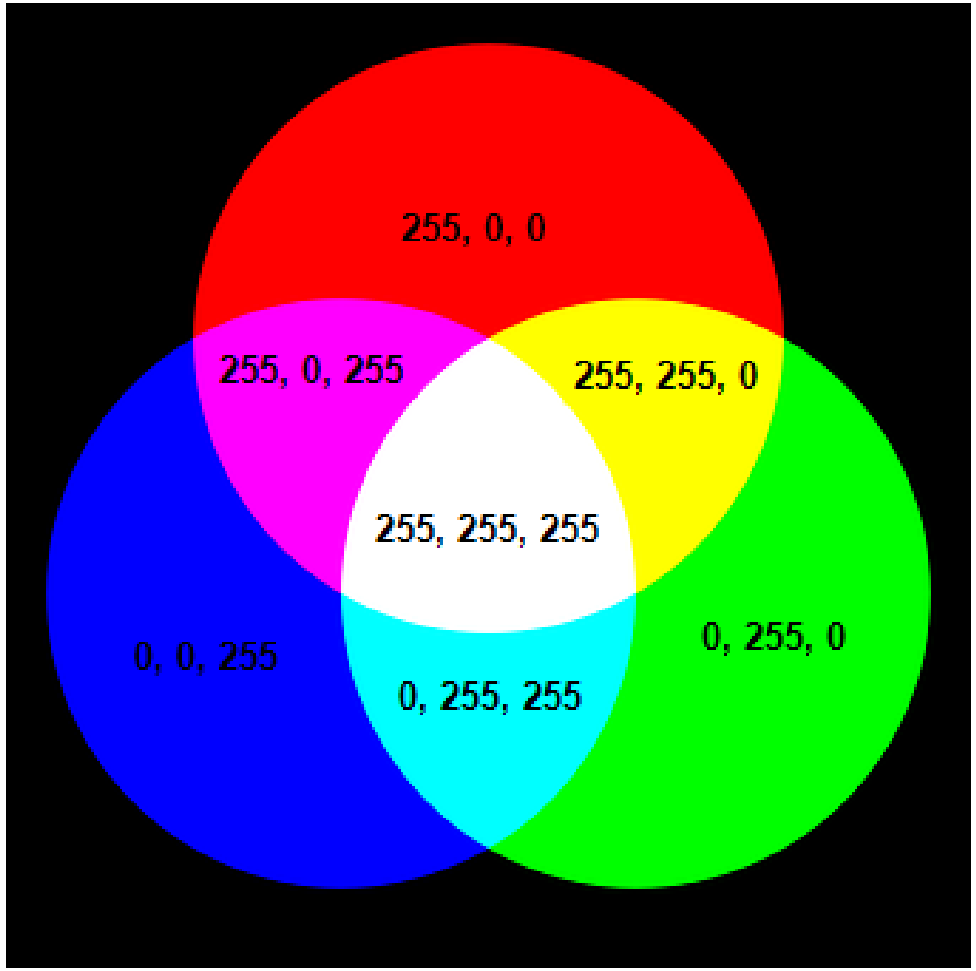
A different type of image is volumetric or 3D images. The number of dimensions in 3D images is equal to three. As a result, we have a \(F(X,Y,Z)\) function. Most of our reasoning still applies, with the only difference being that the triplet \(x_i,y_i,z_i\) is called a voxel (volume element).

These images can be acquired in 3D; that is, the images are acquired in a way that is reconstructed in a 3D space. Examples of such images include medical scans, magnetic resonance, and certain types of microscopes. It is also possible to reconstruct a 3D image from a 2D one.

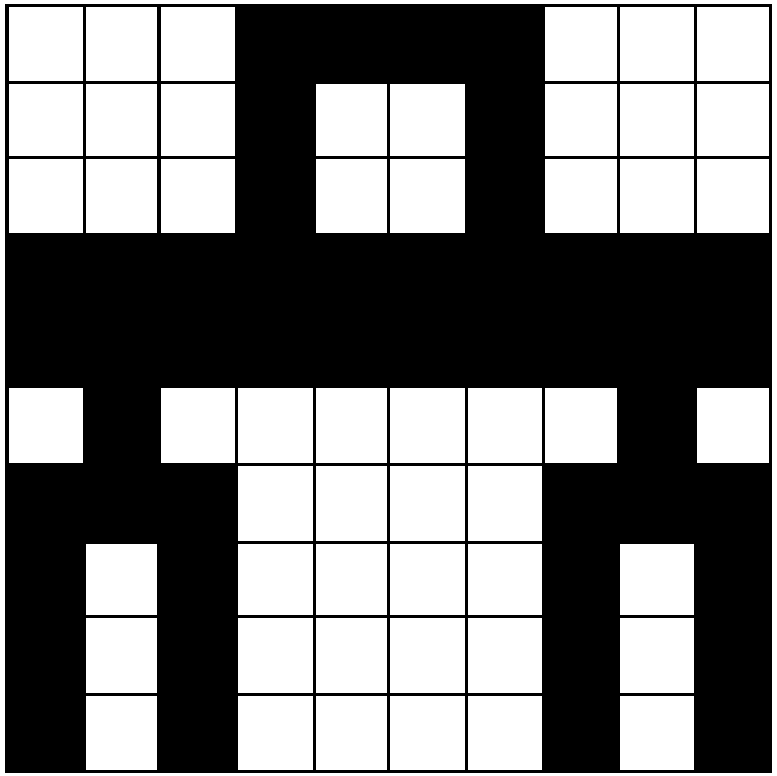
COLOR

- Image channels are simply the different color components that make an image. In reference to the F(x,y), we will have F for each color component.
- Each color has its own intensity level.
- For a channel that picks up the color red, a high intensity means that the color is very red and a low intensity means that there is little to no red in there.

RGB



Binary Images



0	0	0	1	1	1	1	0	0	0
0	0	0	1	0	0	1	0	0	0
0	0	0	1	0	0	1	0	0	0
1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1
0	1	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	1	1	1
1	0	1	0	0	0	0	1	0	1
1	0	1	0	0	0	0	1	0	1
1	0	1	0	0	0	0	1	0	1

There are special types of images where the coordinates F(x_i,y_i) do not describe an intensity value, but instead label a pixel. The simplest example of an operation that results in such an image is separating foreground and background. Everything that is foreground receives the label 1, and everything that is background receives the label 0. These images are commonly referred to as labeled images. When there are only two labels,

But how are images represented in computers? Most commonly by matrices. It is easy to picture an image as a 2D numerical array. This is an advantage because computers handle arrays really well. Seeing matrices as images helps to understand some of the processes in convolution neural networks and in image preprocessing like our example, we call them binary images or masks.

IMAGING

Image acquisition in digital processing is the first step into turning the physical phenomena (what we see in real life), into a digital representation (what we see in our computers). It starts with the interaction between an illumination source and the subject being imaged. This illumination can be of various types, from conventional light sources to more sophisticated forms like electromagnetic or ultrasound energy. The interaction results in energy either being reflected from or transmitted through the objects in the scene. This energy is captured by sensors, which transform one form of energy into another (i.e. transducers converting the incident energy into electrical voltage). The voltage signal is then digitized, resulting in a digital image. To do so, we need advanced technology and precise calibration to ensure that we have an accurate representation of the physical scene. In the next sections, we will explore some of these technologies.

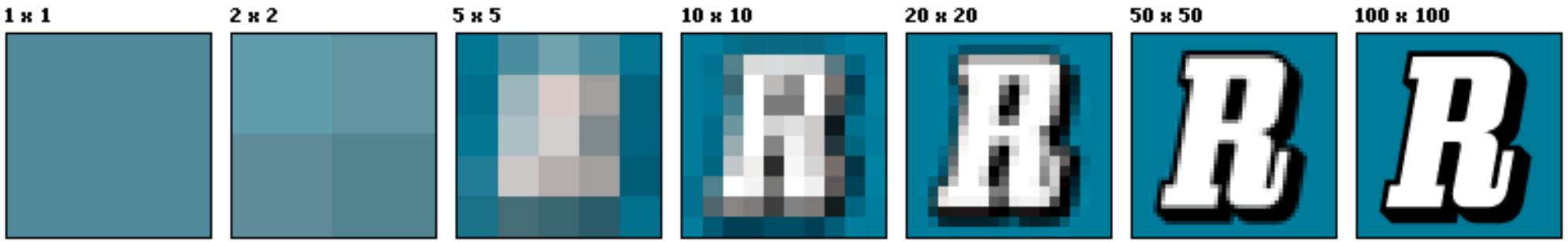
RESOLUTION

➢How much detail an image can represent

- **Spatial resolution** → *detail in space*
- **Intensity (radiometric) resolution** → *detail in brightness*

- 640 × 480 → low resolution
- 1920 × 1080 → higher resolution
- 6000 × 4000 → very high resolution

More pixels = **potentially** more detail



Spatial resolution is the smallest detail you can distinguish in an image.
Can you see individual hairs?
Can you read small text?
Can you distinguish two close objects?

An image is a **grid of pixels**.

Spatial resolution also depends on:
•Lens quality (blur kills resolution)
•Focus
•Motion blur
•Sensor size
•Noise
•Sampling (Nyquist limit)
•----------
What is intensity resolution?
The smallest change in brightness that can be represented
•how smooth gradients look
•how many shades of gray (or color) you can represent

IS MORE MP ALWAYS BETTER ?

Camera A

- 20 MP
- Cheap lens
- 8-bit depth
- Noisy sensor

Camera B

- 12 MP
- High quality lens
- 12-bit depth
- low noise

❌ “More megapixels = better image”
✅ Not necessarily

❌ “Resolution is only about size”
✅ No it’s about **detail you can actually distinguish**

WHAT IS COMPUTER VISION?

the science and technology of making machines see.

It involves the development of theoretical and algorithmic methods to acquire, process, analyze, and understand visual data, and to use this information to produce meaningful representations, descriptions, and interpretations of the world (Forsyth & Ponce, Computer Vision: A Modern Approach)

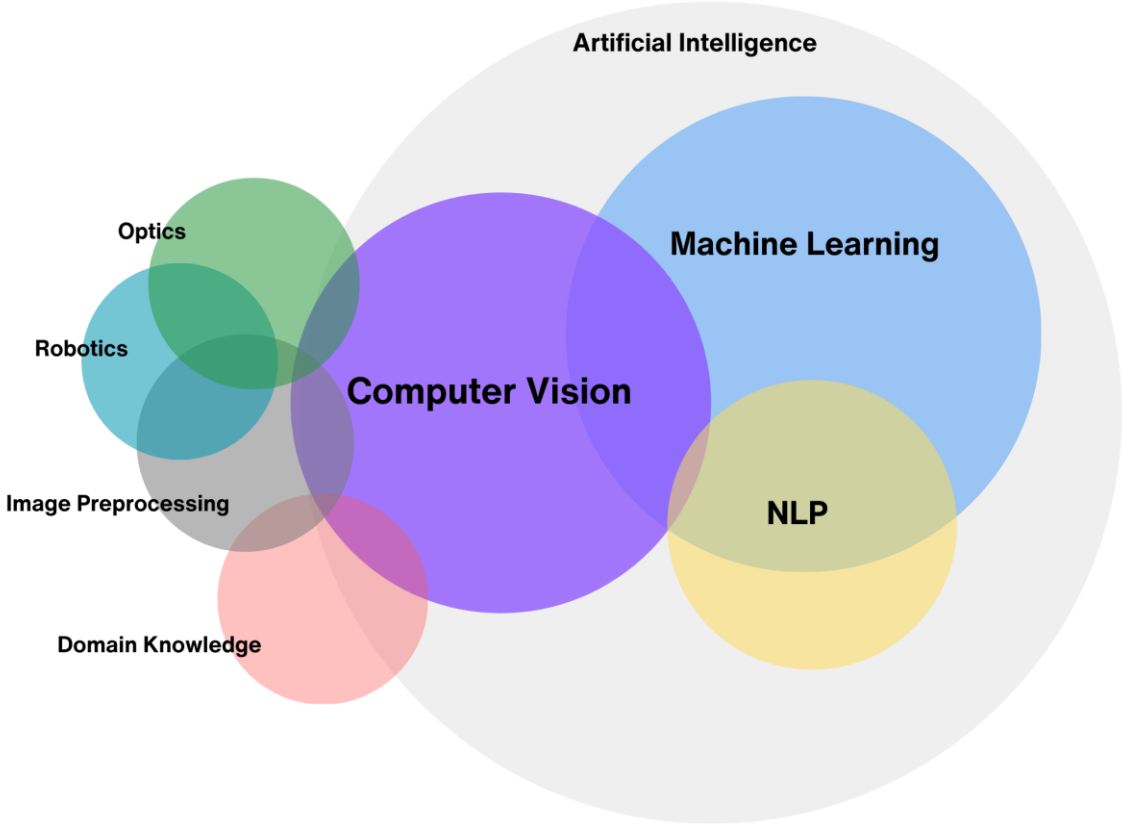


IMAGE UNDERSTANDING

the process of making sense of the content of an image

- **Low-level processes** are primitive operations on images (i.e. image sharpening, changing the contrast). The input and the output are images.
- **Mid-level processes** include segmentation, description of objects, and object classification. The information is an image, but the result is attributes associated with the image. This could be done with a combination of image preprocessing and ML algorithms.
- **High-level processes** include making sense of the entirety of an image, i.e., recognition of a given object, scene reconstruction, and image-to-text. These are tasks typically associated with human cognition.

COMPUTER VISION TASKS OVERVIEW

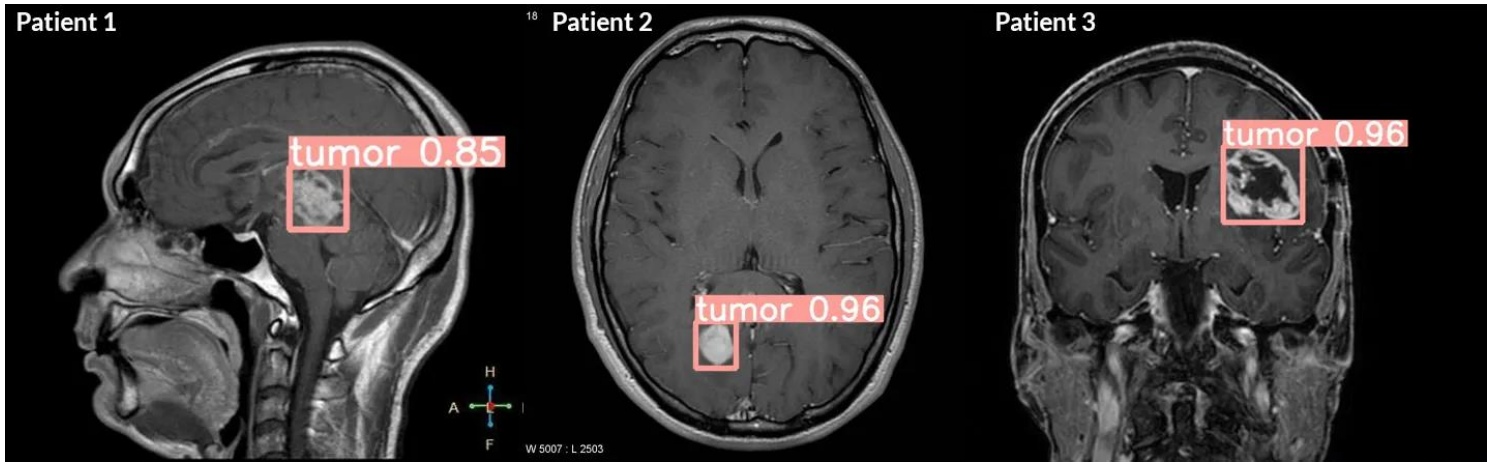
- Scene Recognition
- Object Recognition
- Object Detection
- Segmentation (instance, semantic)
- Tracking
- Dynamic Environment Adaptation
- Path Planning
- Image Captioning
- Image Classification
- Image Description
- Anomaly Detection
- Image Generation
- Image Restoration
- Autonomous Exploration
- Localization

APPLICATIONS OF COMPUTER VISION

- **Retail and E-commerce**
  - By analyzing images or videos of products, computer vision algorithms can identify items, understand their features, and recommend similar products to customers. For example, platforms like Amazon, eBay, and Walmart use computer vision to suggest related products based on users' viewing or purchasing.

- **Autonomous vehicles**

- **Medical Image Analysis**
  - Diagnostic Assistance
  - Segmentation and Detection
  - Treatment Planning and Monitoring
  - AI-Assisted Radiology
  - Drug Development and Research



In today's world, computer vision systems perform ever more challenging tasks. Some of which are even difficult for humans to do. Let's consider India. Do you know India has the highest number of registered two-wheelers in the world? With that many drivers, some of them forget to use a helmet. This is dangerous practice, and can cause severe injuries. To address this issue, the government of India, in collaboration with other institutions, developed a computer vision system that automatically catches riders without a helmet and their license plate. The system imposes strict fines against them to discourage people from breaking the law.

Diagnostic Assistance: Computer vision aids in diagnosing diseases and conditions by analyzing medical images. For instance, in radiology, algorithms can detect abnormalities such as tumors and fractures in X-rays or MRIs. These systems assist healthcare professionals by highlighting areas of concern or providing quantitative data that helps decision-making.

Segmentation and Detection: Medical image analysis involves segmenting and detecting specific structures or anomalies within the images. This process helps isolate organs, tissues, or pathologies for closer examination. For example, in cancer detection, computer vision algorithms can segment and analyze tumors from MRI or CT scans, assisting in treatment planning and monitoring.

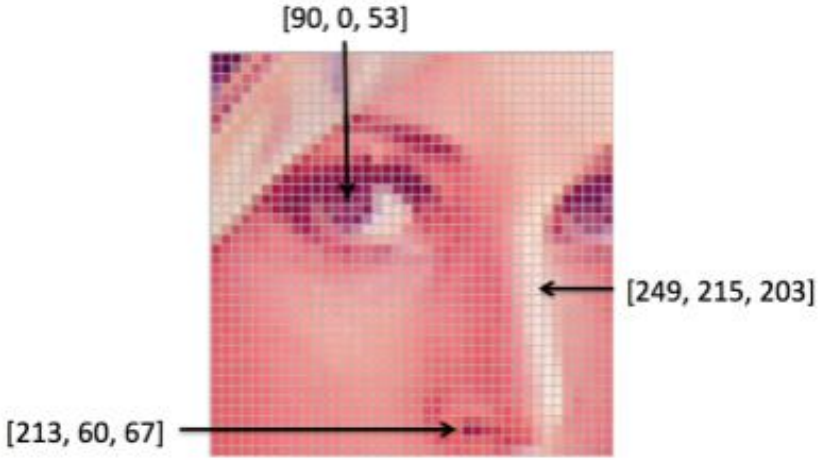
Treatment Planning and Monitoring: Computer vision contributes to treatment planning by providing precise measurements, tracking changes over time, and assisting in surgical planning. It helps doctors understand the extent and progression of a disease, enabling them to plan and adjust treatment strategies accordingly. Doctors were already capable of doing most of these tasks, but they needed to do them by hand. CV systems can do it automatically, which frees us doctors to do other tasks.

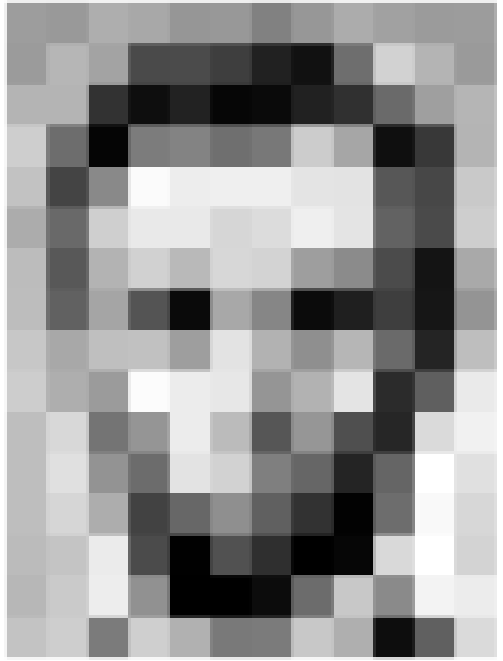
AI-Assisted Radiology: AI-powered systems in radiology assist radiologists by automating routine tasks, reducing workload, and improving accuracy. These systems can flag potentially abnormal findings, provide quantitative analysis, and even predict potential health issues based on patterns identified in medical images.

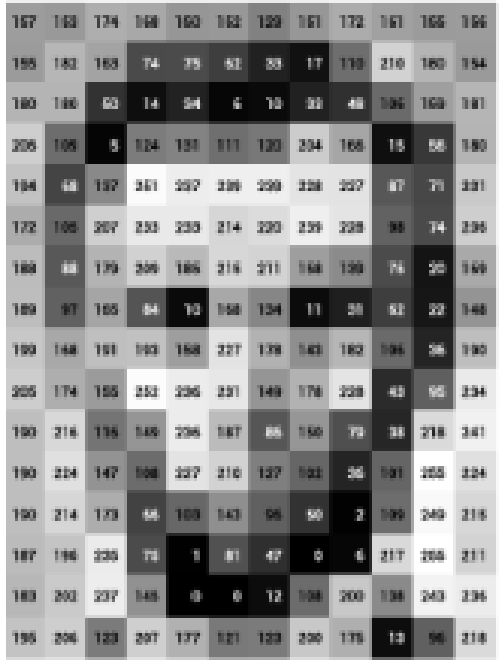
Drug Development and Research: In drug development and medical research, computer vision techniques assist in analyzing cellular structures, tissue samples, or genetic materials. This aids in understanding diseases at a microscopic level, contributing to the development of new drugs, therapies, or diagnostic tools.

FILTERING

$$f(x,y) = \begin{bmatrix} r(x,y) \\ g(x,y) \\ b(x,y) \end{bmatrix}$$







197	193	174	168	153	162	129	161	172	161	155	156
195	182	163	74	75	62	33	17	110	210	180	154
180	186	53	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

MOVING AVERAGE FILTER

- replaces each pixel with the average pixel value of it and a neighborhood window of adjacent pixels

$F[x, y]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$G[x, y]$

	0	10	20	30					

NOTEBOOK 1.FILTERS

FEATURE DESCRIPTION

Features are attributes of the instances learned by the model to be later used to recognize new instances.

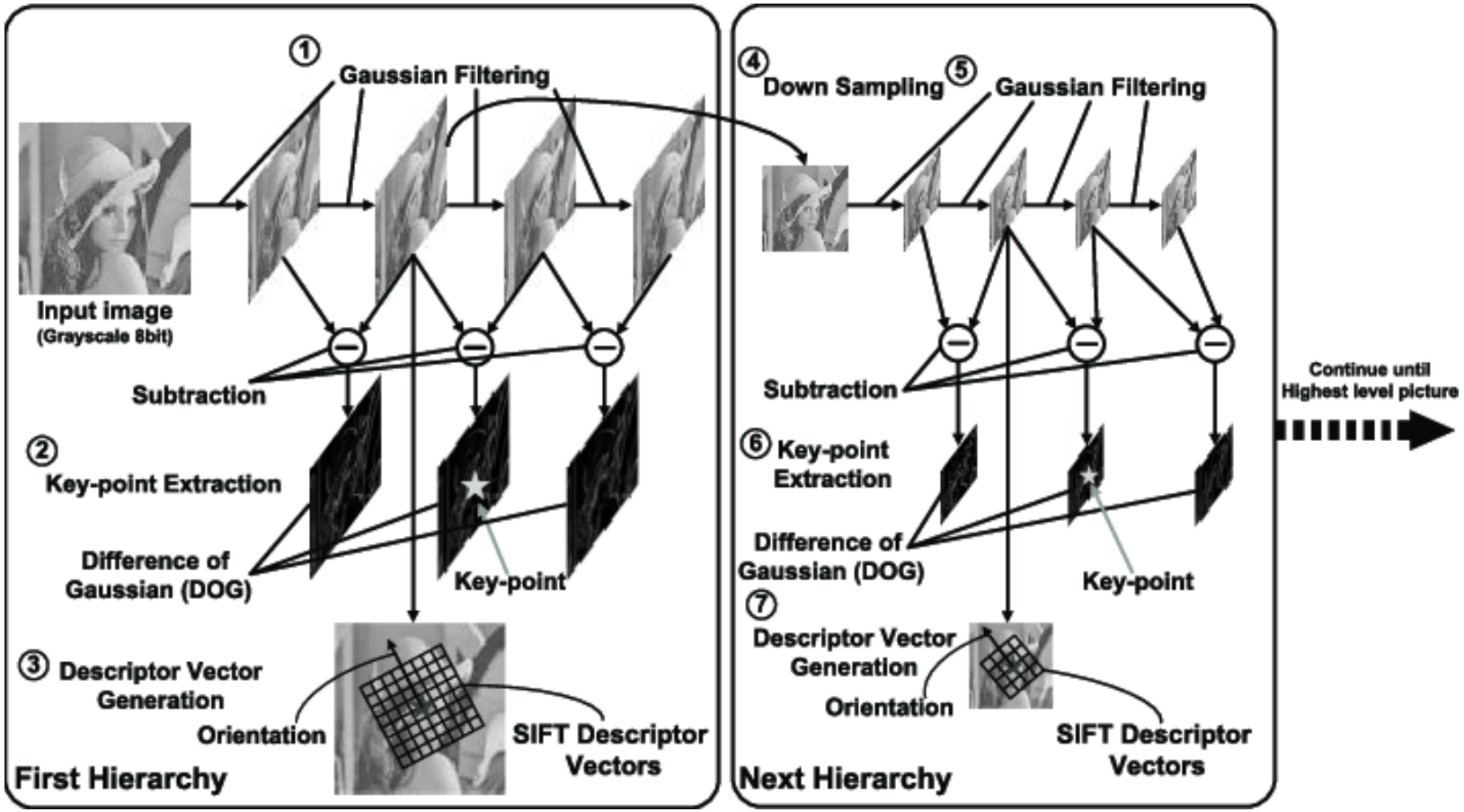
- **Numerical features**
  - Arrays/Lists: Simplest form to store numerical values. Each element in the array corresponds to a feature.
  - Tensors: Multidimensional arrays often used in the machine learning frameworks to handle large sets of numerical data efficiently.
- **Categorical features**
  - Dictionaries/Lists: Assigning categories to numerical labels or directly storing categorical values.
  - One hot encoding: Transforming categorical variables into binary vectors where each bit represents a category.
- **Image features**
  - Pixel Values: Strong pixel values matrices or multi-dimensional arrays.
  - Convolutional Neural Network features(CNN): Extracting features using pre-trained CNN models.

A GOOD DESCRIPTOR ?

- **Invariant to transformation:** Descriptors should ideally be robust to variations like rotation, translation, scaling and changes in illumination. This means that regardless of how an object is positioned or how the image is altered, the descriptor should remain relatively unchanged and contain the same description as the original one.
- **Distinctiveness:** A good descriptor captures unique information about the object. It should be able to discriminate between different objects or parts of an image and be distinct enough to differentiate them from similar elements.
- **Dimensionality:** Good descriptors often have a manageable size, conveying enough information without being excessively large. Balancing dimensionality is crucial for efficiency in processing and storage.
- **Locality:** Descriptors often identify local features within an image. Local descriptors focus on specific regions or keypoints and describe the characteristics of these areas, enabling matching and recognition of similar regions across different images.
- **Repeatability:** Descriptors should be consistent and reproducible across multiple instances of the same object or scene even in the presence of noise or minor variations.
- **Compatibility with Matching Algorithms:** Descriptors are often used in conjunction with matching algorithms to find correspondences between different images. A good descriptor should be suitable for the matching algorithm being used, whether it's based on distance metrics, machine learning models, or other techniques.
- **Computational Efficiency:** Efficiency is really a crucial part of a descriptor, especially in real-time applications. Descriptors should be computationally feasible for quick processing, particularly in scenarios where speed is crucial, such as robotics or autonomous vehicles.
- **Adaptability:** Descriptors that can adapt or learn from the data they are processing can be highly effective, especially in situations where the characteristics of the objects or scenes may change over time. This can increase the useability of the descriptor.
- **Noise Robustness:** Descriptors should be able to handle noise in the image data without significantly compromising their ability to represent the underlying features accurately.

SOME OF THE TECHNIQUES USED IN FEATURE DESCRIPTORS

- **SIFT** (scale invariant feature transform)



- **Scale Space Extrema detection:** It starts by detecting potential interest points in an image across multiple scales. It looks for locations in the image where the difference of Gaussian function reaches a maximum or minimum over space and scale. These keypoint locations are considered stable under various scale changes.
- **Keypoint Localization:** Once potential keypoints are identified, SIFT refines their positions to sub-pixel accuracy and discards low-contrast keypoints and keypoints on edges to ensure accurate localization.
- **Orientation Assignment:** SIFT computes a dominant orientation for each keypoint based on local image gradient directions. This step makes the descriptor invariant to image rotation.
- **Descriptor Generation:** A descriptor is computed for each keypoint region, capturing information about the local image gradients near the keypoint. This descriptor is a compact representation that encapsulates the key characteristics of the image patch surrounding the keypoint. Describing the keypoints as a high dimensional vector.
- **Descriptor Matching:** Finally, these descriptors are used for matching keypoints between different images. The descriptors from one image are compared to those in another image to find correspondences.

FEATURE MATCHING

How can we match detected features from one image to another?

- Brute-Force with SIFT
- Local Feature matching with Transformers (LoFTR)

Imagine you have a giant box of puzzle pieces, and you're trying to find a specific piece that fits into your puzzle. This is similar to searching for matching features in images. Instead of having any special strategy, you decide to check every piece, one by one until you find the right one. This straightforward method is a brute-force search. The advantage of brute force is its simplicity. You don't need any special tricks – just patience. However, it can be time-consuming, especially if there are a lot of pieces to check. In the context of feature matching, this brute force approach is akin to comparing every pixel in one image to every pixel in another to see if they match. It's exhaustive and it might take a lot of time, especially for large images.

LOCAL FEATURE MATCHING WITH TRANSFORMERS (LOFTR)

Uses a learning-based approach to feature matching.

Let's keep it simple and use our puzzle example once again. Instead of simply comparing images pixel by pixel, LoFTR looks for specific key points, or features, in each image. It's like identifying the corners and edges of each puzzle piece. And just as someone really good a putting together a puzzle might focus on distinctive marks, LoFTR identifies these unique points in one image. These could be key landmarks or structures that stand out. As we have already learned, it is important that the matching algorithm handles changes in rotation or scale. If a feature is turned or resized, LoFTR would still recognize it. It's like solving puzzles where pieces may be flipped or adjusted. As LoFTR matches features, it assigns a similarity score to indicate how well the features align. Higher scores mean better matches. It's like giving a grade to how well one puzzle piece fits with another. LoFTR is also invariant to certain transformations, meaning it can handle variations in lighting, angle, or perspective. This is crucial when dealing with images that might be photographed under different conditions. LoFTR's ability to robustly match features makes it valuable for tasks like image stitching, where you combine multiple images seamlessly by identifying and connecting common features.

REAL-WORLD APPLICATIONS OF FEATURE EXTRACTION IN COMPUTER VISION

- **Facial Recognition**
  - **Security Systems**: Airports and public spaces
  - **Consumer Electronics**: Smartphones like the iPhone (Face ID)
  - **Healthcare**: Facial recognition aids in diagnosing genetic conditions.
  - **Marketing and Retail**: Companies use facial recognition to gauge customer reactions to products or advertisements, adapting strategies based on emotional responses.
- **Object Tracking**
  - **Automotive Safety**: Tesla's Autopilot system.
  - **Sports Broadcasting**: Hawk-Eye technology in sports like tennis and cricket tracks ball movement, aiding in accurate decision-making.
  - **Wildlife Conservation**: Camera traps equipped with object tracking algorithms help monitor animal populations and movements.
- **Anomaly Detection**
  - **Public Safety**: In urban surveillance, anomaly detection algorithms help identify suspicious activities or left-behind objects.
  - **Industrial Quality Control**: For instance, BMW uses computer vision to detect minute defects in car parts during production.
  - **Healthcare Diagnostics**: Anomaly detection aids in identifying tumors or other abnormalities.

NOTEBOOK 2.FEATURE MATCHING

APPENDIX

- Sampling : which digitizes the coordinate values.
- Quantization : which converts amplitude values into discrete quantities.

The resolution and quality of a digital image significantly depend on the following:

- The number of samples and discrete intensity levels used.
- The dynamic range of the imaging system, which is the ratio of the maximum measurable intensity to the minimum detectable intensity. This also plays a crucial role in the appearance and contrast of the image.

PRE-PROCESSING FOR COMPUTER VISION TASKS

- **Operations in Digital Image Processing**
  - Logical
  - Statistical
  - Geometrical
  - Mathematical
  - Transform operations